

EDI — the grand daddy of electronic commerce

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Electronic data interchange (EDI) has been around for more than a decade. It addresses a fundamental and enduring business problem — the exchange of data between the computer systems of organisations. It is currently the cornerstone of a number of successful global companies. However, its future lies with new ‘trusted third parties’ which combine data management with public key infrastructure, offering enhanced message switching and message translation in response to market trends.

‘eBusiness’ has a number of forms — business to consumer, now growing fast as ‘Web-shopping’, consumer to consumer, and business to business, the longest established. In the last category there is a major sub-area of ‘supply-chain-driven’ trading. This was the first area of automation and was initiated by the process-driven industries such as retail distribution, automotive manufacture and international logistics. Indeed, whole industries are today totally dependent on EDI-driven processes.

1. EDI today and its history

Electronic business, what’s new!? By 1992 BT had already published a case study [1] in the use of EDI, having just completed the initial roll-out of EDI trading relationships with 25 of its key suppliers. This constituted exchanges of 500-600 orders per day, representing 60% of those from BT’s procurement system, and an order value of £20M per month. Even earlier, in 1987, a pilot EDI system had been implemented with William Sugdens Ltd of Wakefield, manufacturer of BT corporate clothing. This system had been based on an embryonic Telecom Gold EDI service, and once operational it substituted for the monthly delivery of paper orders — two articulated lorries full, which were manually emptied by wheel-barrow.

These were some of the first examples of a completely integrated supply chain between BT and its suppliers, and they dramatically accelerated the fulfilment process for

internal customer orders. Why? — for the same operational and strategic reasons that persist today (see section 1.2 below), which is why EDI remains a good thing to do.

1.1 Definition of EDI

So what exactly is EDI? Electronic data interchange is the exchange of structured business data between the computer systems of trading partners in an agreed standard format. It is analogous to e-mail except that transactions take place directly between computer systems instead of human beings — and it is for this reason that the information needs to be more rigorously structured. In most instances a value added network (VAN) service acts as intermediary between two (or more) trading partners, offering store/collect messaging, audit trail, and resolution of communications protocol differences (Fig 1). These

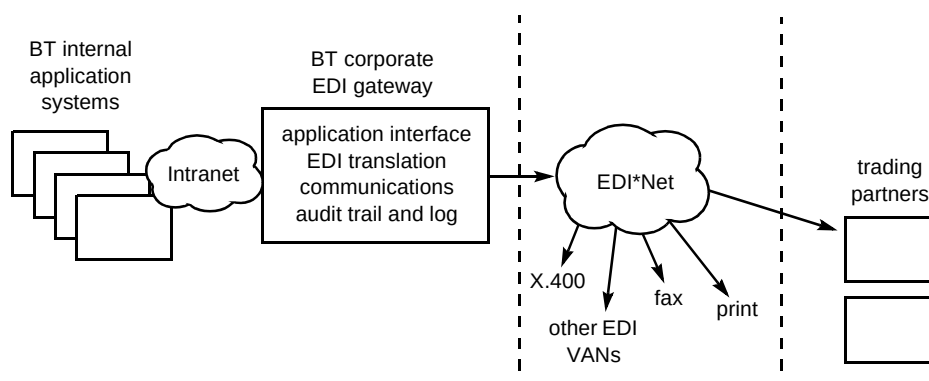


Fig 1 Electronic data interchange (EDI) process model.

Interchanges may contain many different document or message types. The interchange and message structures are defined by national or international standards such as TRADACOM (UK Retail), ANSI X.12 (US), or the global UN/EDIFACT (ISO 9735) standard [2]. The latter has published over 200 message types which are constructed from a common set of 'building blocks' or segments (Fig 2).

1.2 The business problem — advantages of EDI

The fundamental issue that EDI addresses is the need for businesses to exchange documents, such as orders and invoices, without human intervention. The proposition is obviously attractive to large corporations which need to both centralise their operations and reduce the vast amounts of paper they handle, but it can be equally attractive to smaller enterprises; for example, the suppliers of BT Charge Cards are relatively small businesses, and yet receive millions of orders from BT each year. The key to adoption is the transaction volume, not the value of the underlying trades.

Strategic benefits

- Faster trading cycle — particularly important where speed and accuracy of the ordering/invoicing system is vital. It has been estimated that 80% of computer

output is re-keyed into another computer with human error rate 1—5% and latency of 4—5 days.

- Just-in-time manufacturing — refers to the ability to produce minimal sized batches of finished goods only when needed, i.e. responding to market pull; in an extended supply chain it leads to minimal stock holdings by all parties, and hence reduced operating costs.
- Dictating terms of trade — in a competitive market, the ability to offer significant cuts in delivery time may be a considerable competitive advantage.

Operational benefits

- Reduced costs — EDI enables minimal stocks to be held with resultant savings in the costs of storage, insurance, warehousing and security. Improved audit trails lead to better stock management and accountability. Reduction in manual processing (re-keying, filing, etc) enable the reallocation of staff to more beneficial areas.
- Improved cash flow — vast improvements in matching invoices against orders and receipts become possible leading to payment on time. There is a direct correlation between reduced stock handlings and cash-flow.

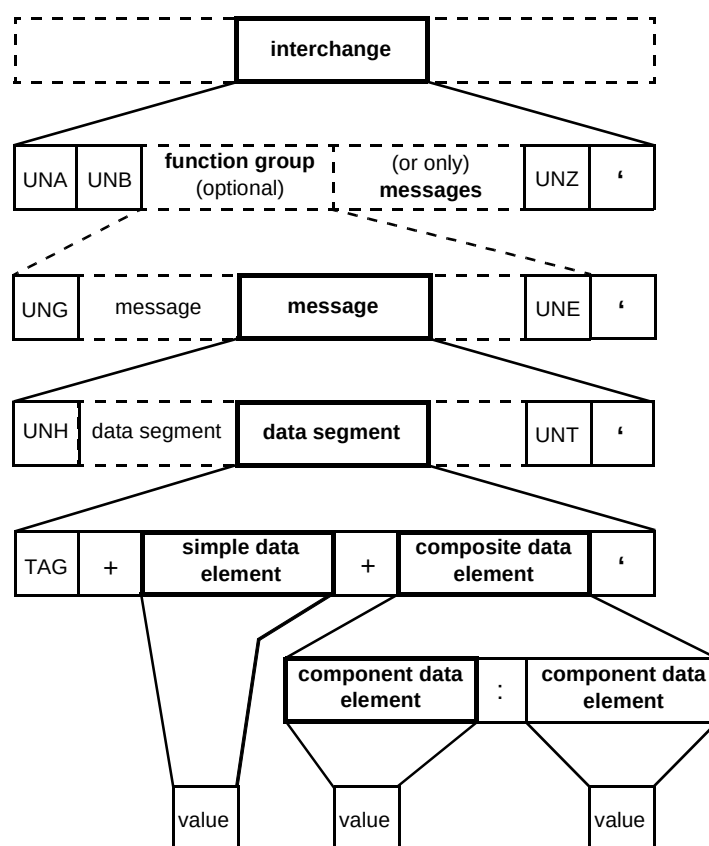


Fig 2 ISO 9735 electronic data interchange for administration, commerce and transport (EDIFACT) application level syntax.

- Security and error reduction — data quality is key to getting things right first time, and reducing the cost of failure. A feature of many EDI systems is automatic confirmation of receipt of the electronic document.

1.3 History of the EDI*Net VAN

The EDI*Net product has a long and complex history. In October 1989, BT acquired the Tymnet data networking company from McDonnell Douglas in a deal which substantially increased BT's networking presence in North America (and around the world). In addition to the network, BT also acquired the EDI*Net business, a leading provider of EDI public services in the USA. Following this, BTTymnet-Europe was formed and the UK EDI*Net service launched in 1991. In 1994 EDI*Net was transitioned from BT North America to MCI, who acted as wholesaler to BT-Concert, who in turn distributed EDI*Net systems software to licensors in Korea (KT Net) and Taiwan (TradeVAN) as well as BT Syncordia Systems in the UK. Concert launched a commercial global EDI service in 1997 based on the MCI and BT Syncordia Solutions domestic EDI*Net services, each of whom owned and operated a Tandem host machine running identical software. The EDI*Net software is a proprietary application developed and designed specifically for Tymnet/BT/MCI. Via the Concert network, the Concert EDI service claimed 'virtual' world-wide reach — a key value proposition for the service. Concert is now working jointly with BT's UK EDI*Net product management for future product development and enhancement (i.e. IP and Internet-based technologies for true eBusiness integration). The EDI product and its capabilities should be a cornerstone of BT's future

eBusiness applications offerings for supply chain management-type applications [3].

1.4 Current level of adoption, UK and world-wide — competitors

The UK EDI market is currently valued at £130M (\$200M) rising to an anticipated £500M (\$800M) by 2004. The world market in the same terms is \$2B and \$7.5B. BT has a good share of the UK market, where it competes with companies such as IBM, GEIS, Harbinger and Kewill. This is clearly a sizeable market and one in which BT can compete.

1.5 How EDI is used within BT — automation of its own supply chain

Most large organisations have adopted a strategic approach to implementing EDI. BT is typical in having its own corporate EDI gateway (Fig 1) which acts as a focal point for connecting internal application systems to those of our trading partners in the 'outside' world. There are several reasons for adopting this approach — centralisation of expert knowledge, non-proliferation of EDI-specific software, reduced implementation effort, centralised gathering of statistics, capture of management information, centralised audit trail, reduction in long-term costs (e.g. operational support, software licenses, maintenance and enhancements). BT's corporate EDI gateway has recently been upgraded to version 6.0 of Stirling Software Inc's GENTRAN product as part of its Year 2000 compliance.

Currently 20 different parts of BT's business use EDI in their day-to-day business (Table 1). The largest of these is

Table 1 Use of EDI within BT's own supply chain.

BT internal system	Business area	Transactions	Volume (per annum) (k)	No. TPs	Example trading partners
CSS-BOPS	Telecom Red	Order/Order Response	250	6	Chubb, Modern, ADT
PPI/CWM	Supply Management	Order	280	320	GPT, BICC, Nortel
ATP	Group Finance	Invoices/Rem Advice	200	15	Nortel, Pirelli, BICC
ARD	Call Centres	Order (timesheets)	1	3	Pertemp, Kelly, Select
DRIVE	Group Motor Transport	Order and Invoice	0.3	5	Ford
ORS	One-Bill [4]	EDI OneBill	0.3	7	IBM, Royal Mail, BGas
BBEMS	Building Services	Utility Bills	0.1	6	Scottish, National, Powergen
Empress	Inland Revenue	P6, P45, P46	2	1	Inland Revenue
Hermes	BT Payphones	Order	0.6	1	Irish Express
COSMOSSCSS	National Lottery	Order/Order response	37	1	Racal
CSS	Cashless/Ring Me Free	Order and PIN No letter	5000	2	IDD, PCC
CSS	International Settlements	Invoice	1.2	20	Telstra
PO System	Cellnet	Order/Invoice	1.5	50	Motorola, GPT,
ALIS	ALIS	Orders	6	12	Interflora, Hotels
Noticing	Digout	Notices, orders, invoices	200	10	Contractors, L/Authorities
Meridian	Nortel New Switch	Order	1.5	3	Nortel Dacom, Innings
AMSOFT	Supply Management	Stock/demand info	10 000	5	Raychem, GPT, BICC
CSS	Directories	Phone Book labels	0.5	6	
CSS	Payphones	Repair orders	5	4	Amalgamated Cleaning Services

BT's procurement unit with around 10 million transactions per annum. Here, the initial introduction of EDI led to massive stock reductions and has subsequently been a key enabler for BTUK's 'single warehouse' strategy. This link also supports 'Direct Fulfilment', the channel whereby orders for customer premises equipment are guaranteed next day delivery anywhere in the UK if placed by 5 p.m. The invoice/payment cycle associated with a large proportion of supply management orders (200,000) is also automated. The invoices are received electronically via EDI and automatically matched against the original purchase order; if there are no discrepancies and receipt of the goods has already been confirmed, payment is automatically made via BACS. Beyond the initial order entry, there is no manual processing at all.

Another example of a 'zero-touch' system, but customer-facing rather than supplier-facing, is illustrated by Telecom Red, which is responsible for the wholesale provision of alarm monitoring services (via conventional telephone lines) to various security companies in the UK, such as Modern Alarms, Chubb and Securicor. Until recently orders were sent to Telecom Red on paper forms (in quadruplicate) with a high level of manual input required for each order. This involved the checking of details on the customer service system (CSS), as well as keying the order details into Telecom Red's system and CSS. With a 100% growth per annum the increasing number of orders required exponential growth in clerical staff and office space. Introduction of EDI has eliminated 98% of manual processing by BT, the original order being entered by the customer, passed to BT, and then automatically validated against customer records held in CSS; if successful, the details are entered into the appropriate work-queues on CSS, and the billing and order processing system. Only those orders in error require intervention.

1.6 *Its limitations — has EDI run out of steam?*

Despite 15 years of government and industry advocacy, EDI has attracted only around 100,000 commercial customers world-wide. The key stumbling blocks have been cost and infrastructure — specifically the fact that most EDI users must connect over dedicated, proprietary VANs. The Internet, however, is changing all that.

2. **The future for EDI**

2.1 *Impact of the Internet*

Proponents of the Internet point to two broad advantages of using open networks for EDI traffic.

Firstly, organisations already connected to the Internet can piggy-back EDI on these links, rather than using a separate VAN network for these connections. Secondly, the use of the global and increasingly ubiquitous Internet for

EDI expands the universe of business trading partners — theoretically, to all businesses that are running interoperable EDI applications.

Does all this spell the end of the VANs? Although it is a threat, they have already recognised that the transport mechanism has to be the Internet, and are coming up with solutions for managed EDI services. These are likely to include technical integration, outsourcing, data-management, business consulting, and EDI competence. There is an inertia for keeping business with the well-known VANs, who are able to guarantee service levels and delivery times, while there are lingering security concerns about sending mission-critical business documents over the public Internet.

The key issues facing the EDI VANs are:

- how to make life simpler and cheaper for their customers,
- opening up the small-to-medium sized enterprises market,
- simple WWW-forms for EDI,
- spontaneous business-to-business trade,
- convergence with the Internet, which brings a variety of file transfer protocols (SMTP, FTP, HTTP) and new document standards (e.g. XML).

These are dealt with in the following sections.

2.2 *More standards — OBI, RosettaNet, ICE, XML*

Renewed interest in eBusiness has spawned a number of new industry groups, each with its own standards for commerce — in the US there are as many as twenty-four. RosettaNet [5], for example, recognises that EDI and bar code standards present a foundation for electronic transactions across many industries, but suggest that it is time to complement them with deeper IT industry-specific standards. Like-wise the 'Open Buying on the Internet Consortium' (OBI) [6] are driven by a shared vision to facilitate the rapid implementation of Internet-based electronic commerce (eCommerce) — completing version 1.0 of the OBI standard in March 1998. However, one has to question the need for new standards for common business documents when so much effort has already gone into producing a single global standard, UN/EDIFACT, which already has ISO recognition.

The umbrella for these new standards might turn out to be extensible mark-up language (XML) [7]. XML itself does not define particular business document formats, rather it is a meta-language that defines the syntax for a particular document type definition (DTD). A DTD defines a

particular set of 'tags' and 'tag attributes' for a given document type. The current W3C [8] Recommendation is XML 1.0 (February 1998) and while there has been much talk of mapping EDI standards to XML, there has been less action. EDIFACT embodies 20 years' analysis of business processes and the message flows and data elements could easily be adopted by XML. The EDIFACT standard also lends itself to object-oriented definitions (e.g. 'class' of Name-And-Address {NAD} used for purchaser, delivery point, invoice point).

2.3 New technological scenarios

A number of new scenarios are presented in Fig 3. An obvious starting point for a VAN is to include TCP/IP in its portfolio of network communications protocols (i). The next step might be to create a Web site on a hosting platform such as BTWebWorld [9] or BTConnect-to-Business, where large corporate or 'hub' organisations could publish simple forms for completion by their 'spoke' trading partners. There are two variants of this kind of service, network centric (ii) and client centric (iii). The network-centric model is really a managed data service for SMEs, and an opportunity that VANs are keen to capitalise on as it is an easy extension to their service. The client-centric model allows SMEs to manage their data locally, more easily integrate with existing applications, and work off-line. Either can be built into a standard Web browser, either as simple hypertext mark-up language (HTML) forms, or more sophisticated Java applications. A logical extension of the client-centric model is to by-pass the VANs all together and use other Internet protocols to transmit data to recipients. Scenarios (iv-vi) represent EDI via simple mail

transfer protocol (SMTP) mail, via the World Wide Web, and using file transfer protocol (FTP).

A version of the SMTP model has been commercially available since May 1998 — Harbinger/Premenos' Templar product combines standard EDI formats with MIME e-mail and public-key encryption from RSA Data Security Inc, which is used to establish non-repudiation of origin and receipt through the use of digital signatures. Commercenet [10] are pushing for a 'standard' EDI via MIME system so that all vendors' software interoperates, and are offering a certification process (at a cost) to support it. Ultimately, existing VANS could be supplanted by new trusted third parties which combine data management with a public key infrastructure (PKI), offering enhanced message switching and message translation.

Direct point-to-point transfers, such as the FTP model, while most economical, require the establishment of more sophisticated gatekeeper firewalls. However, point-to-point transfers have historically been used for high-speed and/or real-time eCommerce (e.g. travel), but this requires both ends to be active and to date has always been conducted over private networks. On the Internet it requires a whole new set of security/access control procedures to prevent unauthorised access.

2.4 Web EDI — a solution for the 'hub-spoke' paradigm?

While classical EDI has addressed the need for bulk data transfer between large corporate or 'hub' organisations, it has always proved rather too expensive for the smaller

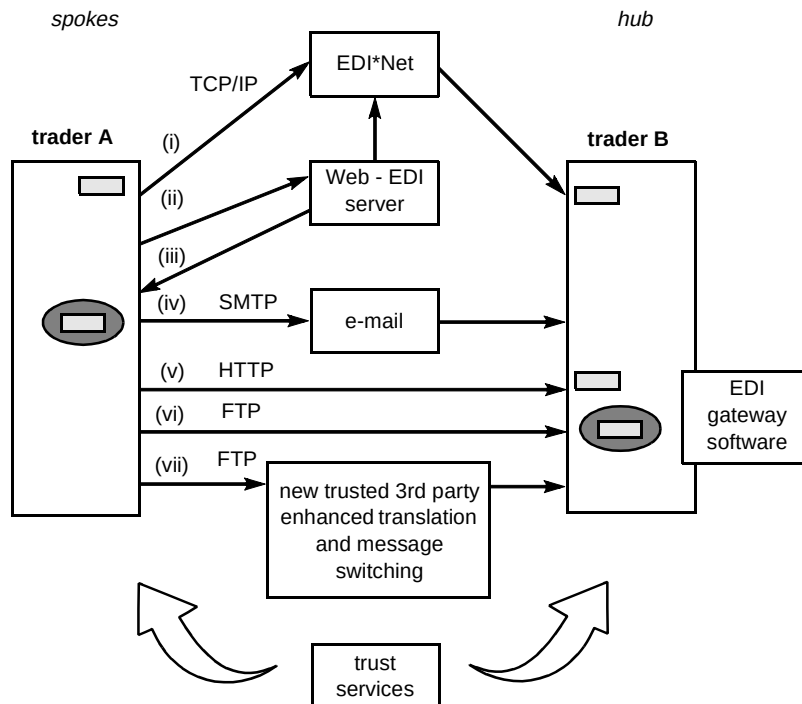


Fig 3 EDI over the Internet.

'spokes', who generally act as suppliers to the corporates. Indeed the adoption of EDI by 'spokes' has been less through choice and more through terms of trade dictated by the 'hubs'. Web-EDI is seen by many as a logical next step for EDI and is in part the answer to this imbalance. It allows companies to set up Web sites that display pages to their trading partners and capture business data (e.g. orders) on-line. An EDI message is then created from the data and passed to their existing EDI processing system. The service can either be operated by the VAN, which is an important business opportunity, or by the companies themselves using EDI tool-kits now available from several software vendors. However, while it is easy to capture data coming into the network/company, it is more difficult to provide data in the reverse direction, especially when there may be many hundreds of trading partners. To resolve these issues is a challenge as it requires either real-time connection to the host system within the company, or a secure intermediate storage system (or the use of hybrid technologies, i.e. Web page for data input, e-mail or fax for output). As yet none of these solutions are ideal as each has deficiencies or high costs. Also, they do not offer true application-to-application interfacing as a person has to key in data to the Web pages — an act which may duplicate their system's data and possibly not retain a copy for them.

2.5 Setting up spontaneous commercial relationships

The 'retail' (or 'business-to-consumer') model of Web-based trading is generally how people interpret eCommerce. It is characterised by World Wide Web catalogues and shopping malls, and a few companies, Amazon Books [11] for example, have become high profile as a result. This business-to-consumer model is spontaneous, using session-level security protocols, such as secure sockets layer (SSL) and secure hypertext transfer protocol (S-HTTP). The customer just asks for goods, offers a payment method (credit-card or a cyber-cash), the goods are supplied (in the case of 'soft' goods immediately), the payment settled, and the customer may never be seen again. While this model is suitable for retail trading, in the business-to-business world the norm is for longer term trading relationships to be established, rather than being 'built on the fly' during a network session. The process involves tendering, credit and quality checks, the placement of a contract to supply goods over a period of time, and then repeat order/invoice/payment cycles.

Mechanisms to support what could be called 'ad hoc EDI' have yet to be fully implemented. However, there is a move to more fluid relationships where a company may invite quotes/tenders for the supply of goods by publishing their requirements on the Internet. In theory, at least, any company in the world could respond and if successful might proceed to set up a new trading relationship immediately. New methods of validating, on-line, that the vendor exists, and that they are who they claim to be, are needed. Similarly the vendor needs to establish the legitimacy of the

buyer. This can all be supported by digital signatures that are certified by recognised organisations (governments, banks, etc) as part of national/global PKI. Once the veracity of the vendor has been established, then actual (electronic) trading can take place. Agreed data standards will need to exist for this (which is where EDI started). The order/invoice/delivery cycle can all be exchanged using EDI/XML supported by PKI, and will dominate the high-volume exchange of documents between businesses and multiple trading partners. These transactions will include customs and shipping needs, both of which are big users of EDI. The actual payment can be achieved by sending electronic payment requests via the SWIFT network just like BACS is used today.

3. Conclusions

EDI may be the grand-daddy of eCommerce, but with age comes wisdom. Many 'new' eCommerce products have already come and gone and the future of others is far from certain. EDI addresses a fundamental and enduring business problem — the exchange of data between the computer systems of businesses. For larger companies this requirement is unlikely ever to disappear and so, in one form or another, the future of EDI is guaranteed. Astute players in the eCommerce market-place recognise this and are investing appropriately, indeed some (GEIS [12]) have developed complete businesses around their core EDI products. Establishing spontaneous business-to-business trading relationships is the real challenge. It will be the competitive edge of the future, and the prevalent means of trading in the global village of eCommerce.

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Ian Jones joined BT in 1987 after graduating from Cardiff University with a BSc in Computer Systems and working for a small specialist software company. During his time in BT he has worked in various fields of software development including supply management systems where he first worked on EDI. He was closely involved with the introduction of BT's Corporate EDI Gateway and interfaces to many internal systems including the One-Bill billing system, ordering and invoicing for supply management and the National Lottery. More recently he has worked on Internet-related projects including BTInternet and the BT WWW-EDI Trading Station. He continues to participate in standards groups both nationally and internationally, monitoring the potential for EDI over the Internet.